Difference Bounds Matrix: how to compute the normal form?

- The normal form can be computed by the **Floyd-Warshall algorithm**
  - N iterations indexed by $k = 0, 1, \ldots, N-1$
  - Iteration k derives b_{ij}^{k+1} defined as the shortest path from τ_j to τ_i that does not visit any intermediate node with index larger than k
    - **Initialization**: b_{ij}^0 is the initial value of coefficient b_{ij}
    - **First iteration** ($k = 0$): b_{ij}^1 is the shortest path from τ_j to τ_i that does not visit any intermediate node with index larger than 0 0 é l'indice del nodo
    - **Last iteration** ($k = N-1$): b_{ij}^N is the shortest path from τ_j to τ_i

Algorithm 2 Floyd-Warshall algorithm

```
for each i, j ∈ {0, 1, ..., N − 1} ∪ {⋆} with i ≠ j do
    b_ij^0 = b_ij
end for
for each k = 0, 1, ..., N − 1 do
    for each i ∈ {0, 1, ..., N − 1} ∪ {⋆} do
        for each j ∈ {0, 1, ..., N − 1} ∪ {⋆} with i ≠ j do
            b_ij^(k+1) = min{b_ij^k, b_ik^k + b_kj^k}
        end for
    end for
end for
```

∪{⋆} → SI CONSIDERA ANCHE IL GROUND.

→ SI CONSIDERA ANCHE I VAR $b_{i\star}$ o $b_{\star i}$